

Lab 1

10% Overall Grade

This lab requires you to create an AWS key pair, set up and configure an AWS EC2 instance, associate an elastic IP address, establish a remote development environment using VSCode Remote SSH, install Linux packages, install Anaconda, install the Apache web server, and execute the genomic analysis pipeline script.

Part A: Create an AWS Key Pair for Secure Access

1. Log in to the AWS Management Console and navigate to the EC2 Dashboard.
2. Create a new key pair with a name that includes your name (e.g., FIRST_NAME_Lab_1_Key).
3. Download the key pair file (a .pem file) and store it in a secure location on your local machine.
4. Screenshot Requirements:
 - Key pair created in the console.
 - Downloaded Key file (don't need to open, just visible in file explorer).

Part B: AWS EC2 Instance Configuration and Deployment

1. Server name will be FIRST_NAME_Lab_1_Server
2. Use the Ubuntu 24.04 AMI.
3. Select the instance type t2.large.
4. Assign the key pair you created in Part A.
5. Allow HTTP and HTTPS access.
6. Change the hard drive size from 8gb to 30gb
7. Screenshot Requirements:
 - EC2 Instance launch configurations.
 - EC2 Instance details once launched.

Part C: AWS EC2 Elastic IP association

1. Allocate an AWS EC2 Elastic IP
2. Assign the name FIRST_NAME_eip_lab_1
3. Associate the elastic IP with your server from part B
4. Screenshot Requirements:
 - EC2 Instance details with updated IP address

Part D: Remote Development with VSCode Remote SSH

1. Install the VSCode Remote - SSH extension on your local machine.
2. Configure your SSH settings (using your key pair) to enable VSCode to connect to your EC2 instance.

3. Open VSCode and connect to your EC2 instance using the Remote - SSH extension.
4. Open the home folder (/home/ubuntu) in VSCode
5. Open your integrated terminal in VSCode.
 - Terminal -> New Terminal (in VSCode GUI)
6. Download the genomic analysis pipeline script using the following command (Terminal)
 - git clone https://github.com/SohailHanif/Bioinformatics_Cloud_Computing.git
7. Screenshot Requirements:
 - VSCode window showing the remote connection established.
 - Genomic analysis script file opened in VSCode from the EC2 instance.
 - Git clone command execution in the integrated terminal

Part E: Linux Package Management

1. Update software repositories package index
 - sudo apt update
2. Download any new software package updates
 - sudo apt upgrade -y
3. Verify the tree command is not installed by default
 - tree
4. Install the tree command
 - sudo apt install tree
5. Verify the tree command is now installed
 - tree
6. Screenshot Requirements:
 - Terminal output showing the update and upgrade commands
 - Terminal output showing the installation and execution of the tree command.

Part F: Anaconda and Executing the Genomic Analysis Pipeline Script

1. Download the latest Miniconda installer for Linux using wget
 - wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
2. Run the installer script
 - bash Miniconda3-latest-Linux-x86_64.sh
3. Accept the license terms (yes during interactive installation)
4. Update shell profile to automatically enable conda (yes during interactive installation)
5. Kill and create a new integrated terminal to reload conda configuration
6. Create a new conda environment
 - conda create --prefix ./bioenv
7. Activate new conda env
 - conda activate ./bioenv
8. Add the conda channels
 - conda config --set channel_priority strict
 - conda config --add channels bioconda
 - conda config --prepend channels conda-forge
9. Install the required bioinformatics packages to run the script
 - conda install entrez-direct sra-tools fastqc samtools bwa gatk4 snpEff -y

10. Verify packages installed correctly using the conda list command
 - conda list
11. Change directory to the lab 1 folder
 - cd Bioinformatics_Cloud_Computing/lab_1/
12. Execute the genomic analysis pipeline script
 - bash genomic_analysis_script.sh
13. Screenshot Requirements:
 - Terminal output of the Anaconda installation steps and prompts.
 - conda list command execution showing all programs have been installed.
 - Successful execution of the genomic analysis pipeline script from Github.

Part G: Apache Web Server

1. Copy your IP address from the EC2 instance details (Don't click open address)
2. Paste your IP address in your web browser URL bar
3. Verify you get a "This site cannot be reached" error
4. Install the Apache web server
 - sudo apt install apache2
5. Verify you can load the Apache webserver home page that is hosted on your server
6. Copy and execute the following commands:
 - sudo rm /var/www/html/*
 - sudo cp -r ~/Bioinformatics_Cloud_Computing/lab_1/ /var/www/html/
7. Paste your IP address in your web browser url bar and view the genomic analysis files
8. Screenshot Requirements:
 - "This site cannot be reached" error when loading IP address initially
 - Terminal output Apache2 being successfully installed
 - Apache web server home page after installation and searching for IP address again
 - Genomic analysis pipeline script files visible when accessing your server IP address

Part H: Environment Cleanup

1. Terminate your EC2 instance
2. Disassociate your elastic IP address
3. Screenshot Requirements:
 - Elastic IP dashboard with no associated IP addresses
 - EC2 dashboard with terminated EC2 instance

Files to Submit:

- A PDF summarizing your findings and explanations.
- Screenshots of:
 - Key pair created in the console.
 - Downloaded Key file (don't need to open, just visible in file explorer).
 - EC2 Instance launch configurations.
 - EC2 Instance details once launched.

- EC2 Instance details with updated IP address
- VSCode window showing the remote connection established.
- Genomic Annotation Script File opened in VSCode from the EC2 instance.
- Git clone command execution in the integrated terminal
- Terminal output showing the update and upgrade commands
- Terminal output showing the installation and execution of the tree command.
- Terminal output of the Anaconda installation steps and prompts.
- conda list command execution showing all programs have been installed.
- Successful execution of the genomic analysis pipeline script from Github.
- “This site cannot be reached” error when loading IP address initially
- Terminal output Apache2 being successfully installed
- Apache web server home page after installation and searching for IP address again
- Genomic analysis pipeline script files visible when accessing your server IP address
- Elastic IP dashboard with no associated IP addresses
- EC2 dashboard with terminated EC2 instance